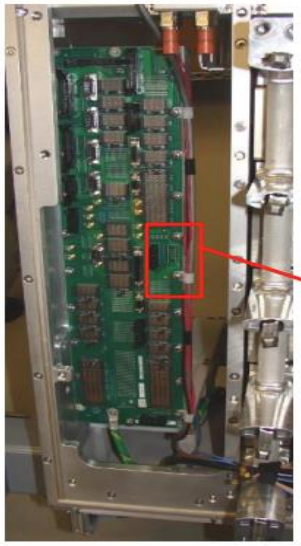
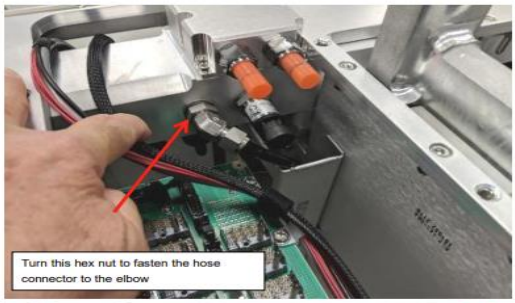
Build Overview:

1 **Support Board Backplane Install**
2 **LED/EMO and Ground Cable Install**
3 **NVRAM PCB**
4 **DB12 Mid-Planes Install**
5 **Disconnect Panel (Dog House) Install**
6 **Install the following cables:**
   - **a.SB25 48V Power Cable**
   - **b.EMO**
   - **c.DSI Power**
   - **d.IEC Power**
   - **e.Scope Panel**
   - **f.TDR Panel**
7 **Pressure Sensor Install7**
8 **Supply QD Assembly Install**



NVRAM memory small board installed state



Turn this hex nut to fasten the hose connector to the elbow

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	638-079-01	PRESSURE SENSOR 0-100PSIG 4-20MA SAE #4 PORT 7/16"-20 UNF	2
2	361-607-08	O-RING 0.351"IDX 0.072"W E0540-80	2
3	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1
4	654-656-01	ELEC-MECH,UFII SP6 SIDEPLANE BOARD ASSY,FOR SEMI TEST	1
5	472-426-16	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 16MM L W/ CONE WASHER. O.D. 11MI	22
6	938-172-01	WI TYWRAP ANCHOR #10SCREWMTG	1
7	835-610-00	HDW CLAMP CBL .312	3
8	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK	1
9	634-157-00	PCBA, ULTRAFLEX II SYS NVRAM, SEMI TEST	1
10	629-403-01	ELEC-MECH, UFII LARGE MIDPLANE ASSY, FOR SEMI TEST	1
11	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	10
12	650-705-01	CABLE UF2 12SLOT SB25 INPUT POWER TELECOM WITH CONN	1
13	813-929-01	HDW VELCRO STRP 5/8	4
14	622-412-01	identification labels (Black THA)	1
15	622-412-00	identification labels (Blue THA)	1
16	651-228-00	ALUMINUM DISCONNECT PANEL BRACKET OR MISC ARTICLE	1
17	406-502-10	SCRW PAN HEAD TORX M5X.8P/10MM SS W/NYLOK	6
18	655-542-02	CABLE UF2 TH INT EMO_LEDS_INTLK SM CONN TELECOM WITH CONN	1
19	637-238-00	CABLE UF2 SPLIT DSI POWER AU CONTACT 48V TELECOM WITH CONN	1
20	938-104-01	4 INCH TIE WRAP BLA	1
21	938-102-00	TIE WRAP 6.5 X 3/16"	2
22	653-832-01	ELEC-MECH DB12 SUPPLY FLEX TUBE W/ QD ASSEMBLY FOR SEMI TEST	1
23	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1

METHOD:

Flow chart of operation steps

BPM preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Materials
↓
Operation Check
↓
Inventory tool

MACHINE:

TOOL NAME	TORQUE VALUE
22mm Long socket wrench	27 N·m
PH2 Phillips torque screwdriver	1.6 N·m
PH2 Phillips torque screwdriver	2.8 N·m
PH2 Phillips torque screwdriver	3.3 N·m
PH2 Phillips torque screwdriver	0.7 N·m
T20 Torx wrench	1.6 N·m
T25 Torx wrench	1.6 N·m
T25 Torx wrench	3.3 N·m
3/16 Socket wrench	0.19 N·m
11/16 open-end wrench	27 N·m
Adjustable wrench	NA
PH1 cross torque screwdriver	0.7 N·m
Flat-blade torque screwdriver	0.7 N•m
PH1 slotted torque screwdriver	0.7 N•m
Paper Tape	NA
Plastic Bag	NA
17mm open end	NA
19mm open end	NA

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

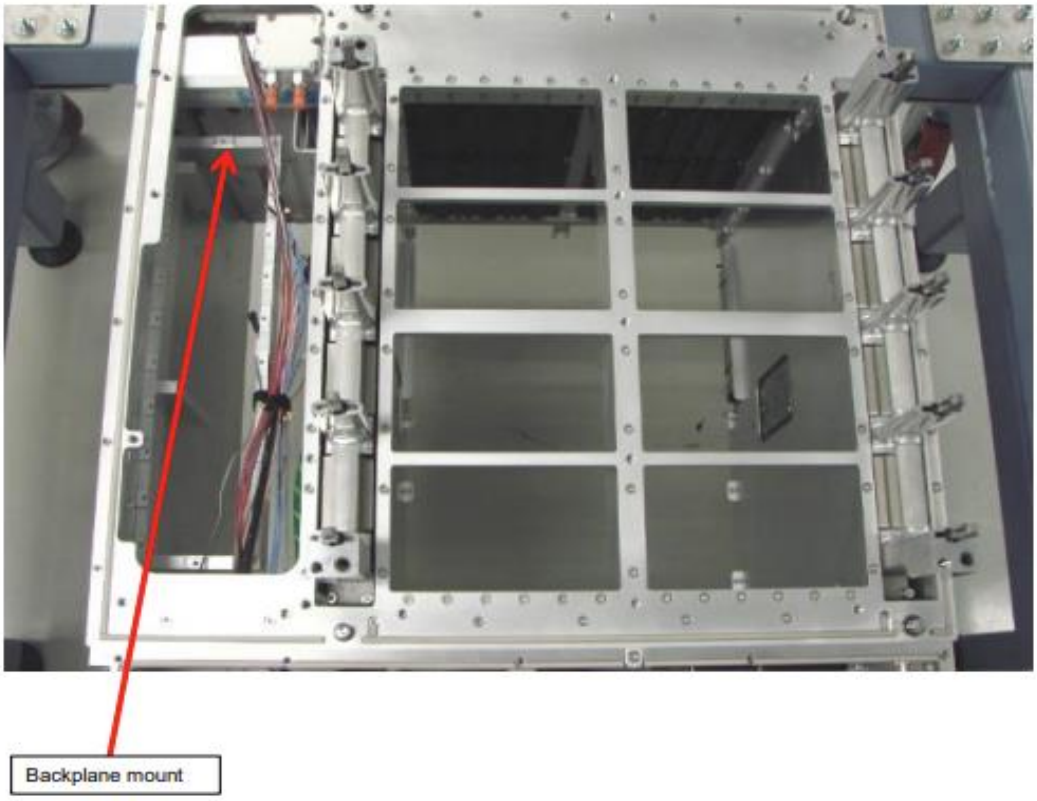
MACHINE:

N/A

METHOD:

Operation steps

1.1 Shake T/H to the position shown in the figure.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

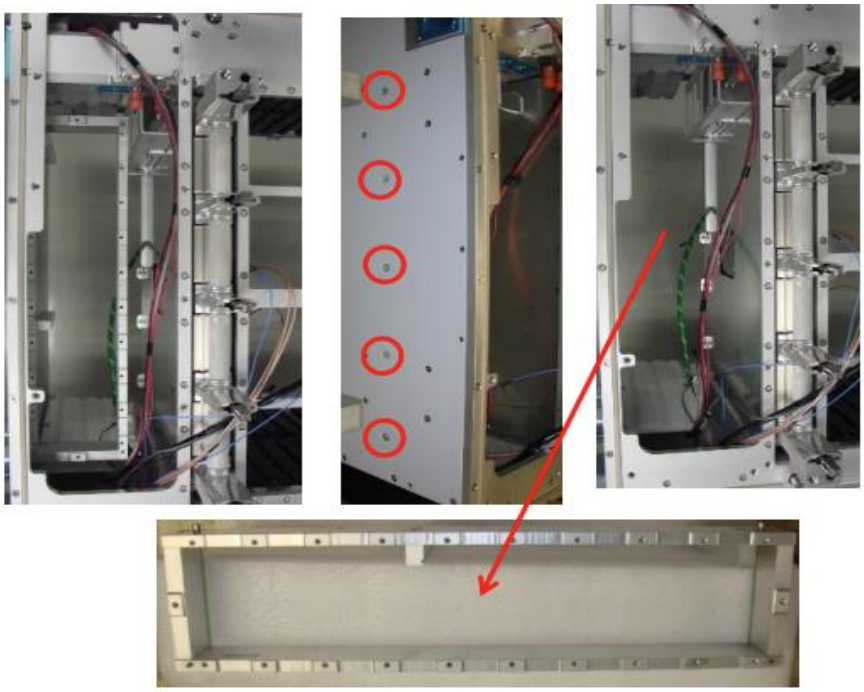
N/A

MACHINE:

4mm Hex Bit, 3.3 N•m

METHOD:

1.2 Loosen the wiring harness from the mounting frame and place the cable on the right side of the supporting base plate area.
with the edge of the bracket.
1.3 Remove 5 M5 screws from the outside of TH as shown in the figure, and then remove the backplane mounting frame from inside the TH.
Save 5 screws and they will be reinstalled later.
Using 4mm Hex Bit with torque value 3.3 N•m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

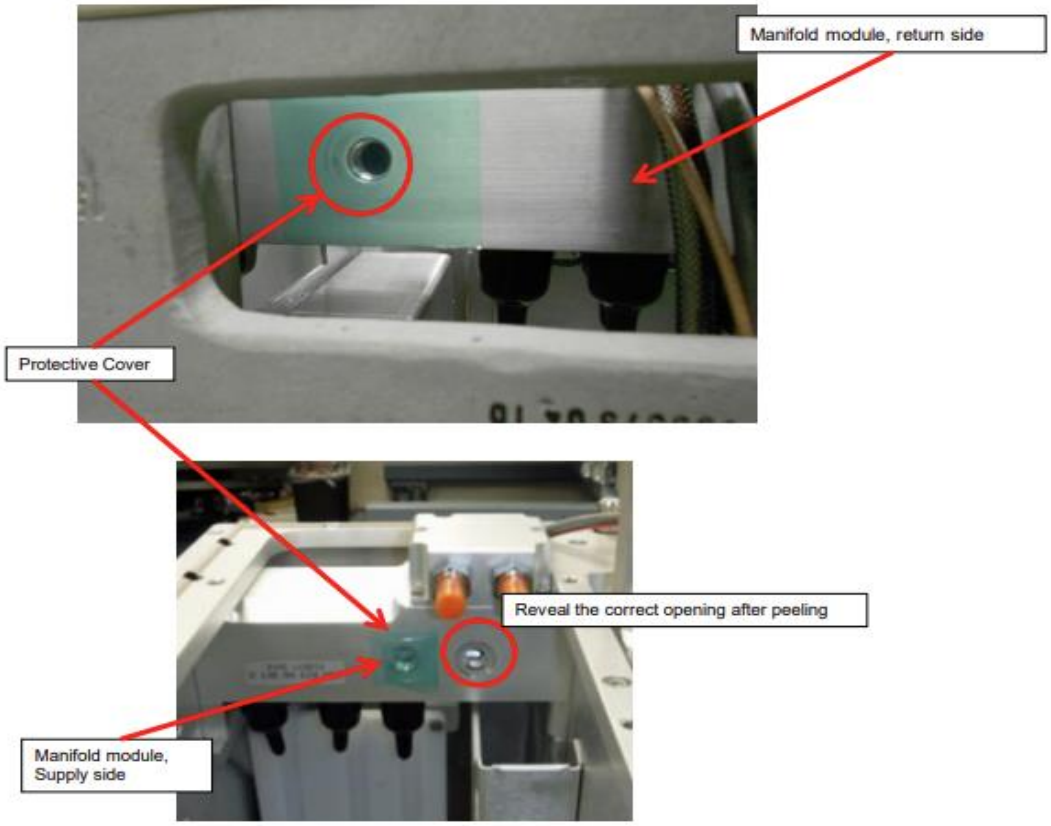
MACHINE:

N/A

METHOD:

2.1 Carefully tear off the plastic protective film on the hole side of the return side of the Manifold module, and also tear off the hole on the right side of the supply side of the Manifold module.

2.2 Check that the outlet, inlet and pressure sensor ports are not damaged or debris remains.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

N/A

METHOD:

Process Aleart: Test Head (Preassure Sensor Assembly)

1. Visual Check O-ring. Ensure No Damaged, No Dented, No Cracked or No Deformation.
2. Ensure installing the O-ring on the pressure sensor and place the O-ring on the bottom of the screw thread (groove).

Figure 1: Good O-ring that has round shape, smooth surface, No dented, No cracked or No deformation.

Figure 2: Good O-ring sit on the bottom of the screw thread (groove).

Figure 3: Reject O-ring that has damaged, dented, cracked or deformation.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

N/A

METHOD:

Process Aleart: Test Head (Preassure Sensor Assembly)

3. Visual Check that the pressure sensor port are not damaged, not broken and no debris.

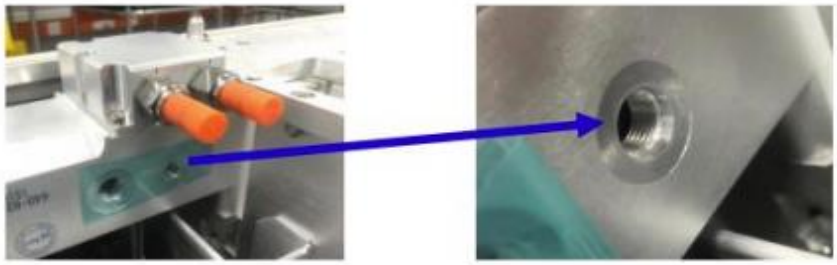


Figure 4: Pressure sensor port in good condition. No damage, No broken, No debris

4. Visual Check the O-ring is flush against the pressure sensor port.

ACCEPT

Figure 5: Good O-ring is flush against the pressure sensor port

MAN:

CERTIFIED OPERATOR, TECHNICIAN

ITEM	PART NUMBER	DESCRIPTION	QTY
1	638-079-01	PRESSURE SENSOR 0-100PSIG 4-20MA SAE #4 PORT 7/16"-20 UNF	2
2	361-607-08	O-RING 0.351"IDX 0.072"W E0540-80	2
3	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1

MACHINE:

22mm long socket wrench 18 N.m. to 20 N·m (Typically 19 N.m.)

METHOD:

2.3 Take 2 pressure sensors 638-079-01.
2.4 Lubricate 2 O-rings 361-607-08 and threads with lubricant 638-030-00.
2.5 Install the O-ring on the pressure sensor thread and turn to the bottom of the thread.
2.6 Install pressure sensors on the air inlet on the return side and the air outlet on the supply side of the Manifold module.
2.7 Tighten the pressure sensor by hand, and then use a 22mm long socket wrench to fix the pressure sensor with a torque value of 18 N.m. to 20 N·m (Typically 19 N.m.)

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	654-656-01	ELEC-MECH,UFII SP6 SIDEPLANE BOARD ASSY,FOR SEMI TEST	1
5	472-426-16	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 16MM L W/ CONE WASH	19

MACHINE:

Glove, PH2 Phillips torque screwdriver 1.6 N•m

METHOD:

3.1 Remove one piece of backplane 654-656-01 from the packaging, and pay attention to wearing ESD gloves when handling the PCB.

3.2 Put the back plate 654-656-01 on the frame according to the hole position of the positioning column as shown in the figure.

3.3 Use 18 472-426-16, M5 screws to lock the back plate to the frame. The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N•m.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
6	938-172-01	WI TYWRAP ANCHOR #10SCREWMTG	1

MACHINE:

PH2 Phillips torque screwdriver 1.6 N•m

METHOD:

3.4 Install a cable clamp 938-172-01 as shown in the figure and fix it with one 472-426-16, M5 screw.
The tool is a PH2 Phillips screwdriver with a torque value of 1.6 N•m.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

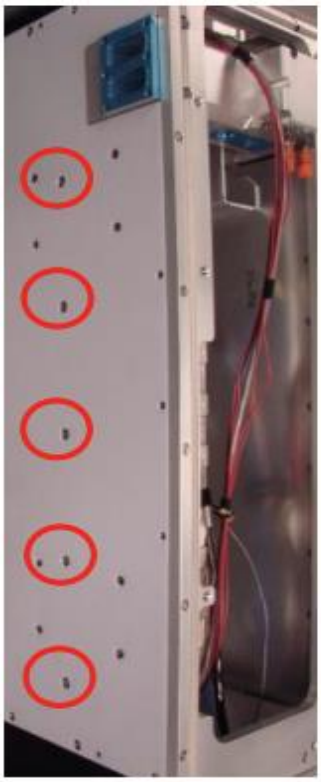
N/A

MACHINE:

N/A

METHOD:

3.5 Reinstall the bracket with the backplane installed on the T/H using the 5 M5 screws previously saved,
The tool is a 4mm Hex Bit with a torque value of 3.3 N•m.
Note: When installing, use the positioning posts on the side of the mounting frame to position the frame on the TH!



MAN:

CERTIFIED OPERATOR, TECHNICIAN

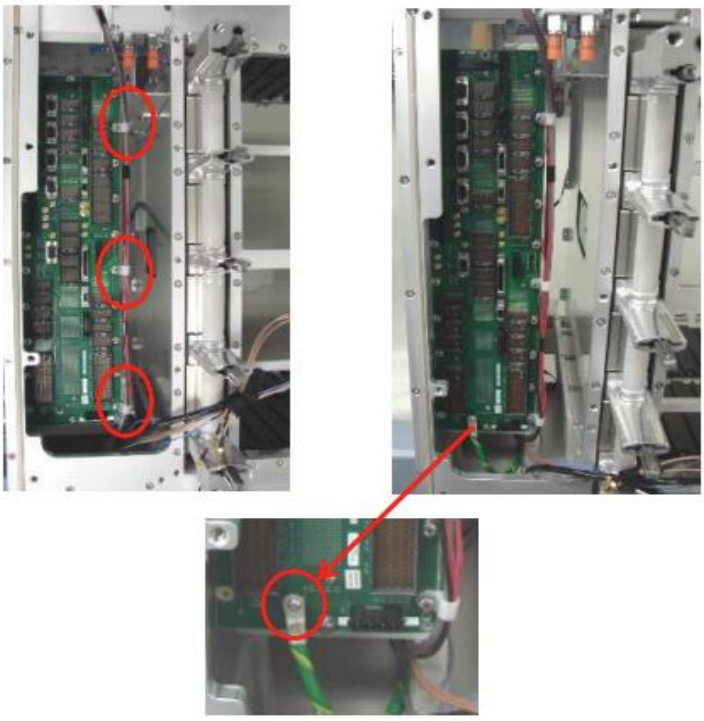
ITEM	PART NUMBER	DESCRIPTION	QTY
7	835-610-00	HDW CLAMP CBL .312	3
8	472-426-16	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 16MM L W/ CONE WASH	3
9	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK	1

MACHINE:

PH2 Phillips torque screwdriver 1.6 N•m, T20 Torx wrench 1.6 N•m

METHOD:

3.6 Use three (835-610-00) clamps to fix the EMO cable and 10G signal cable along the backplane with three M5 screws (472-426-16) as shown in the figure.
The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N•m.
3.7 Use one M4 screw (202-591-06) to lock the ground wire (652-809-00) to the corner of the backplane as shown in the figure.
The tool is a T20 Torx wrench with a torque value of 1.6 N•m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	634-157-00	PCBA, ULTRAFLEX II SYS NVRAM, SEMI TEST	1

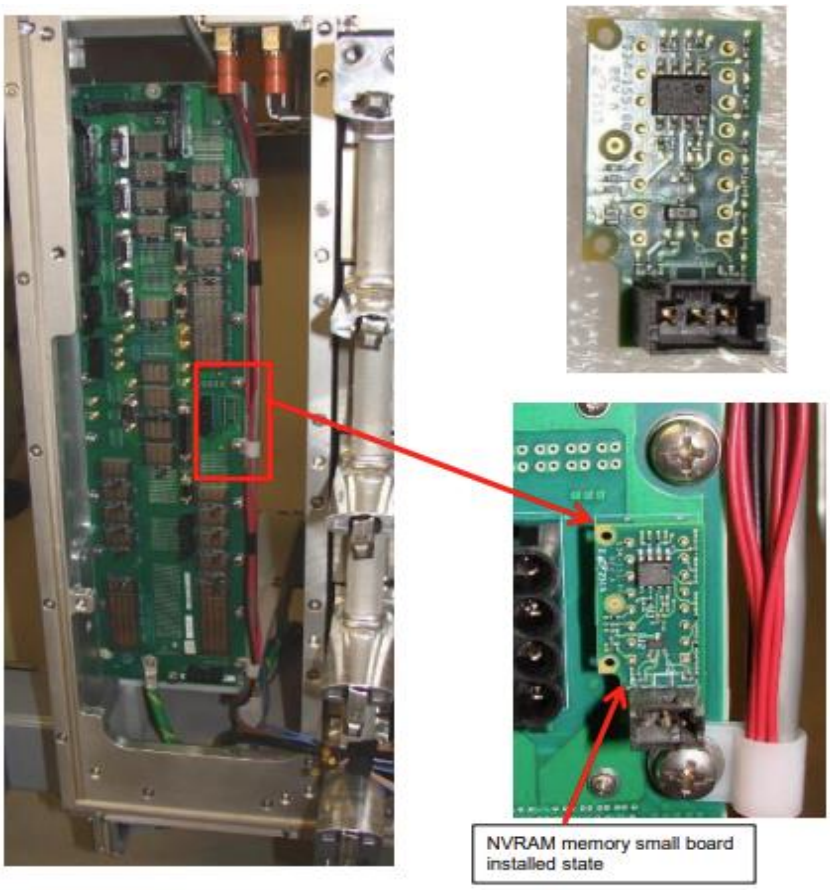
MACHINE:

Glove

METHOD:

4.1 Install a small NVRAM memory board (634-157-00) on the J60 interface of the backplane as shown in the figure, and the direction is as shown in the figure.

Note: Pay attention to ESD protection during installation!



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

3mm Hex Bit torx wrench 1.6 N•m

METHOD:

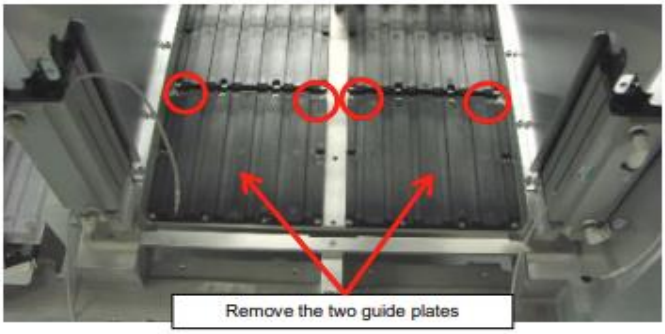
5.1 Shake the T/H to the position shown in the figure (note the position of the cylinder-it should be below the bottom of the T/H).

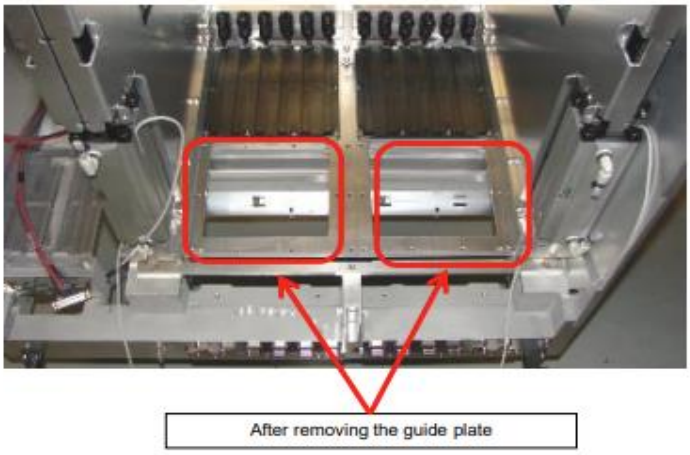
5.2 Use a 3mm Hex Bit torx wrench to remove two black pla

a. Remove the 4 screws circled in the picture (2 for each guide plate), these 4 screws are slightly shorter than others;

b. Remove the remaining 16 screws (8 for each guide plate).

5.3 Save 2 guide plates and 20 screws, place 4 short screws and 16 long screws separately, do not confuse.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

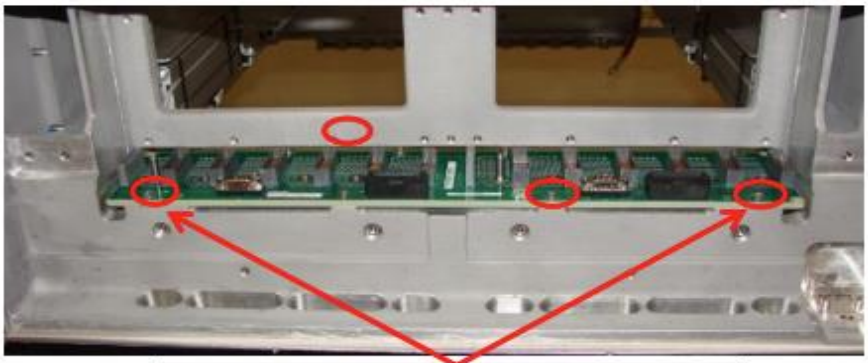
ITEM	PART NUMBER	DESCRIPTION	QTY
11	629-403-01	ELEC-MECH,UFII SP6 SIDEPLANE BOARD ASSY,FOR SEMI TEST	1
12	202-591-10	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 16MM L W/ CONE WASH	4

MACHINE:

Glove, T20 Torx wrench 1.6 N•m

METHOD:

6.1 Shake T/H to the position shown in the figure.
6.2 Remove one piece of MP12 backplane (629-403-01) from the packaging.
Please wear ESD gloves when handling the PCB.
6.3 Put the back plate 629-403-01 on the frame according to the hole position of the positioning column as shown in the figure.
6.4 Use 4 (202-591-10) M4 screws to lock the back plate to the frame.
The tool is a T20 Torx wrench and the torque value is 1.6 N•m.



MP12 is placed according to the positioning column

Safety Tips: N/A

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
13	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	6

MACHINE:

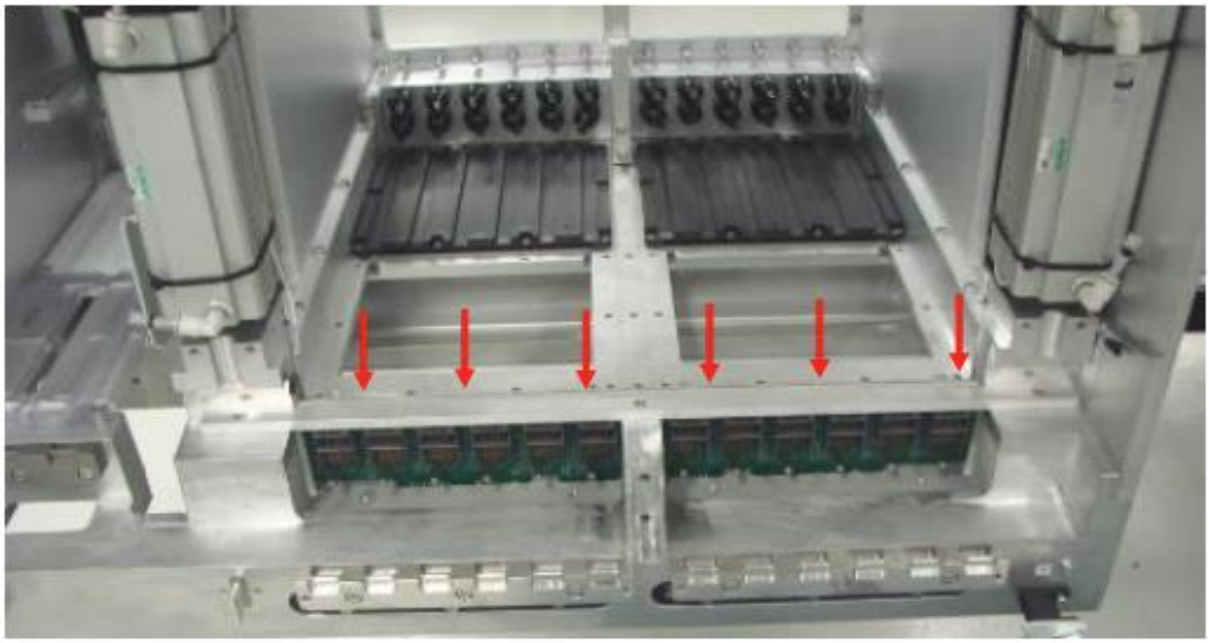
T20 Torx wrench 1.6 N•m

METHOD:

6.5 Shake T/H to the position shown in the figure.

6.6 Use 6 M4 pan head screws (202-591-10) to fix the inner edge of the MP12 to the T/H frame.

The tool is a T20 Torx wrench with a torque value of 1.6 N•m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

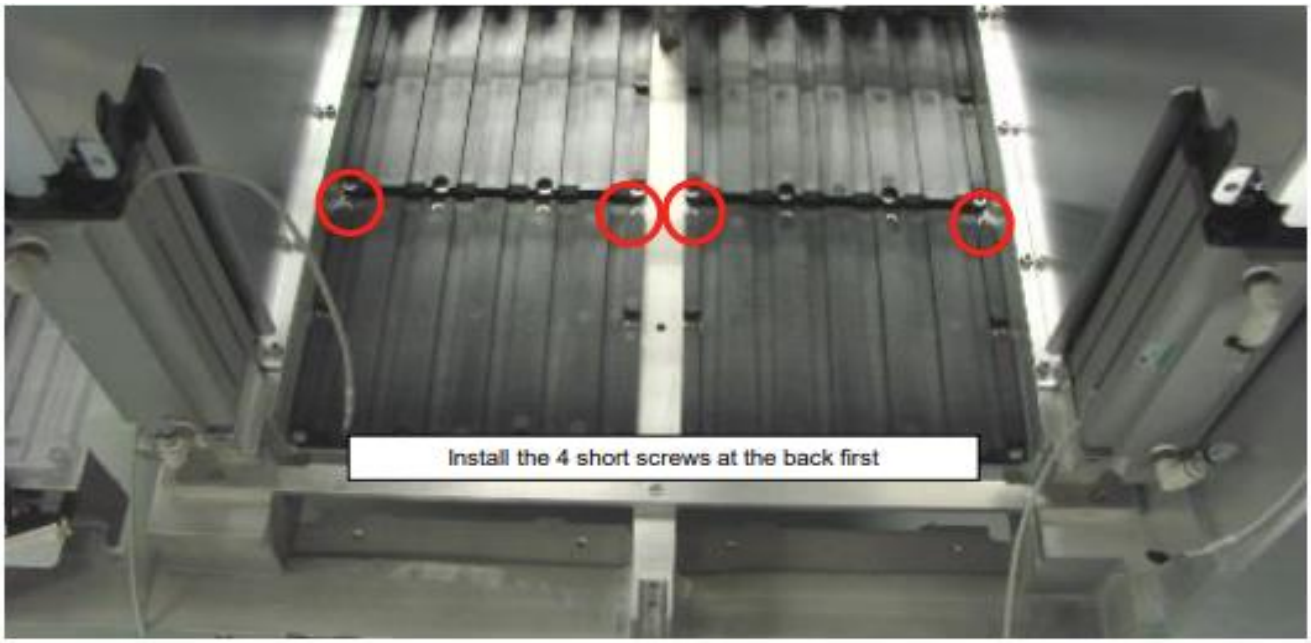
N/A

MACHINE:

3mm Hex Bit Torx wrench 1.6 N•m

METHOD:

6.7 Two guide plates near the door that were removed before reinstallation:
a. First install the 4 short screws circled in the picture (two for each guide plate), the tool is a 3mm Hex Bit Torx wrench, and the torque value is 1.6 N•m;
b. Install the remaining 16 long screws (8 for each guide plate), the tool is a 3mm Hex Bit, and the torque value is 1.6 N•m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

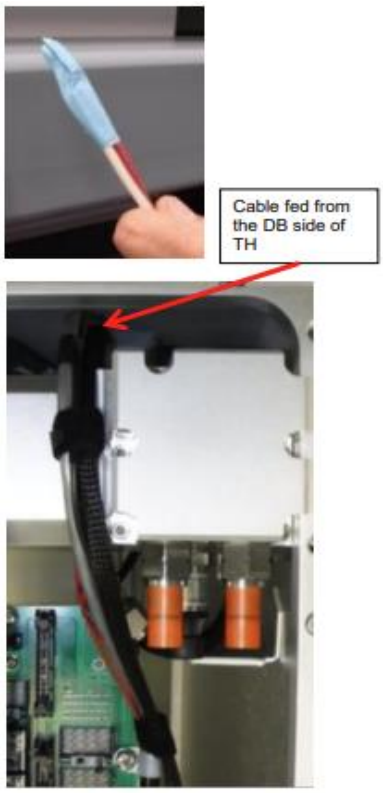
ITEM	PART NUMBER	DESCRIPTION	QTY
14	650-705-01	CABLE UF2 12SLOT SB25 INPUT POWER TELECOM WITH CONN	1
15	813-929-01	HDW VELCRO STRP 5/8	3

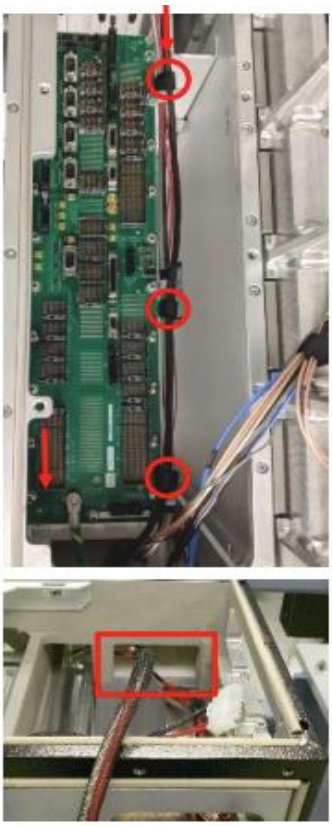
MACHINE:

paper tape

METHOD:

7.1 Take one SB25 48V cable 650-705-01 and wrap the thread with paper tape.
7.2 From the DB side of the TH, feed the wire into the channel and route it along the right side of the backplane, into the opening directly below the backplane. Use three cable ties 813-929-01 to tie the wire as shown in the figure.
7.3 Continue threading to the power supply side of T/H.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
16	622-412-00	identification labels (Blue THA)	1

MACHINE:

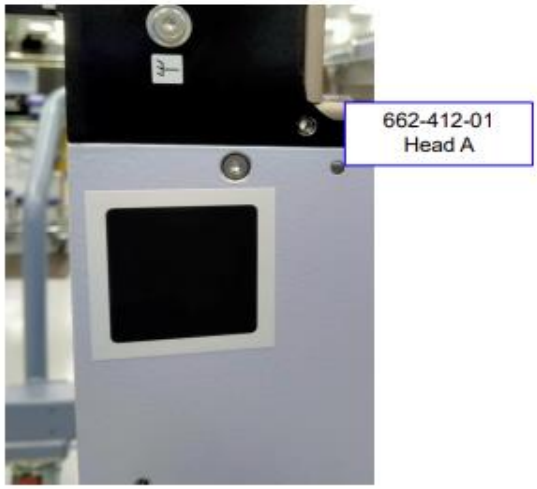
N/A

METHOD:

8.1 Take T/H identification labels: 662-412-01 black THA

662-412-01 Test Head A

8.2 Shake T/H to the maintenance position.
8.3 Tear off the label, align the left side to the T/H panel,
and just under the screw (as shown in Figure 662-412-01).



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

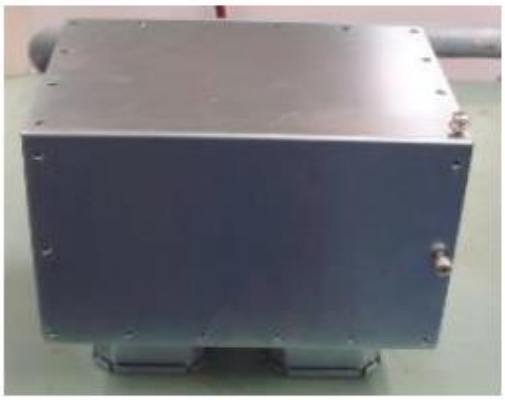
ITEM	PART NUMBER	DESCRIPTION	QTY
17	651-228-00	ALUMINUM DISCONNECT PANEL BRACKET OR MISC ARTICLE	1

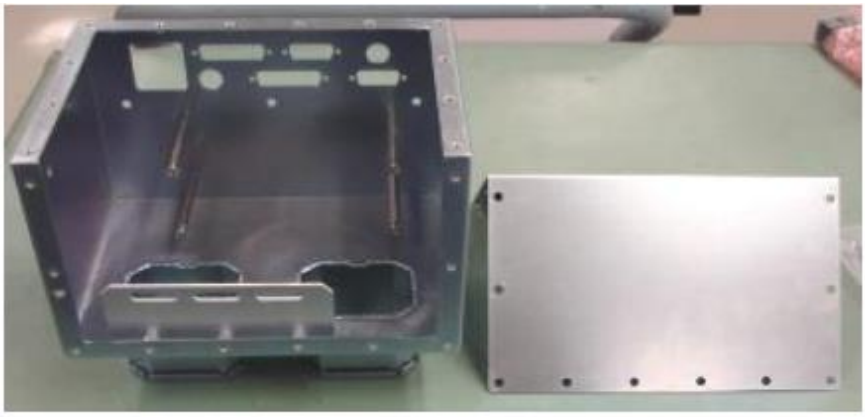
MACHINE:

PH2 Phillips screwdriver, plastic bag

METHOD:

9.1 Take one bulkhead panel 651-228-00.
9.2 Use a PH2 Phillips screwdriver to remove the outer cover of the bulkhead panel.
A total of 22 screws need to be removed, and the screws are wrapped in plastic bags
and attached to the outer cover for future use.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

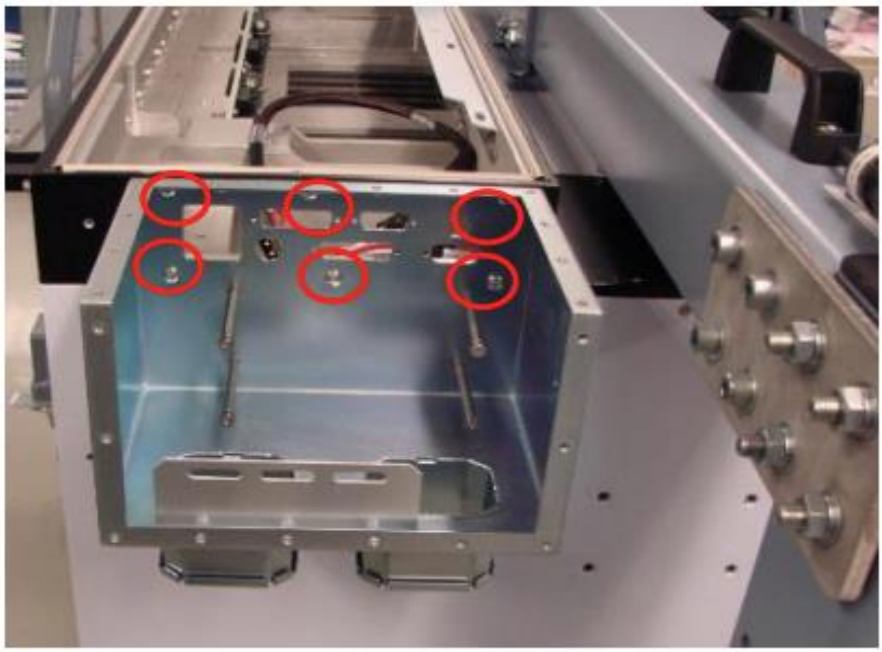
ITEM	PART NUMBER	DESCRIPTION	QTY
18	406-502-10	SCRW PAN HEAD TORX M5X.8P/10MM SS W/NYLOK	6

MACHINE:

T25 Torx wrench 3.3 N•m.

METHOD:

9.3 Install the bulkhead panel to the T/H frame with 6 M5 screws 406-502-10 as shown in the figure,
The tool is a T25 Torx wrench with a torque value of 3.3 N•m.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
19	655-542-02	CABLE UF2 TH INT EMO_LEDS_INTLK SM CONN TELECOM WITH CONN	1

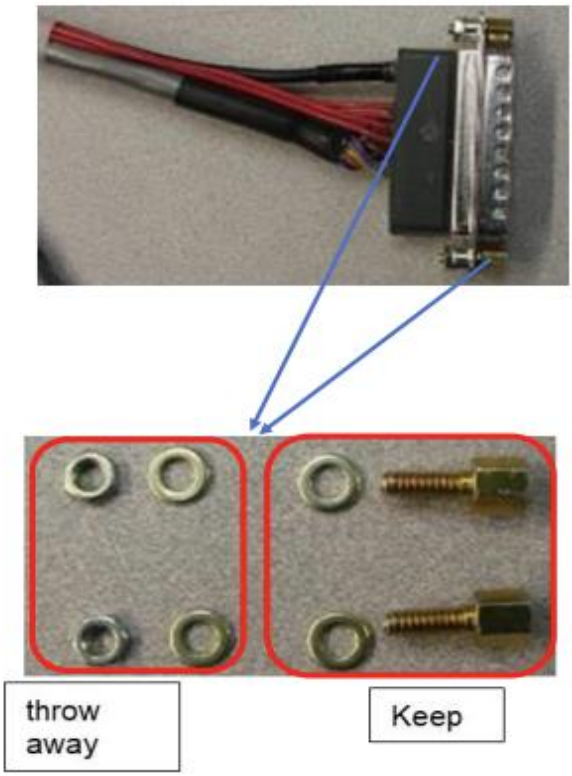
MACHINE:

3/16 socket wrench 0.19 N•m, marker

METHOD:

9.4 Install the EMO line 655-542-02 connector to the bulkhead panel.

a. Remove the screw washer from the connector of the EMO cable, leave 2 studs and 2 flat washers, and discard the rest;



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

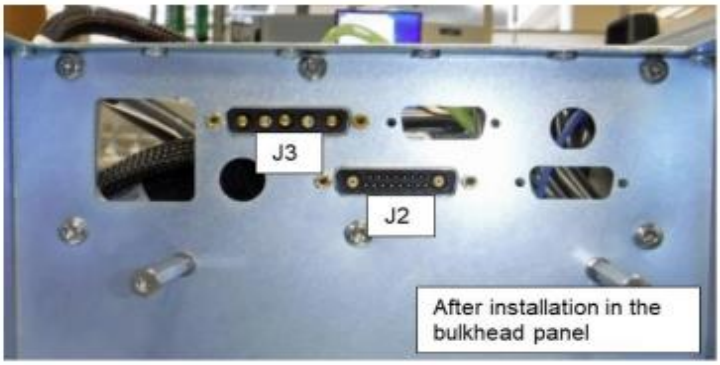
N/A

MACHINE:

N/A

METHOD:

b. Lock the connector labeled "T/H DISCON_PNL J2" of EMO to the "J2" interface of the panel
with the reserved studs & washers,The tool is a 3/16 socket wrench with a torque value of 0.19 N•m.
9.5 Install the SB25 line 650-705-01 connector to the bulkhead panel.
a. Remove the screw washer from the connector of the SB25 cable,
Leave 2 studs and 2 flat washers, and discard the rest;
b. Use the reserved studs & washers to lock the "T/H DISCON_PNL J3" connector
to the "J3" interface of the panel,
The tool is a 3/16 socket wrench with a torque value of 0.19 N•m.
c. As shown in the figure, put the other end marked SB25 J48" through the front opening of the T/H, see next page.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

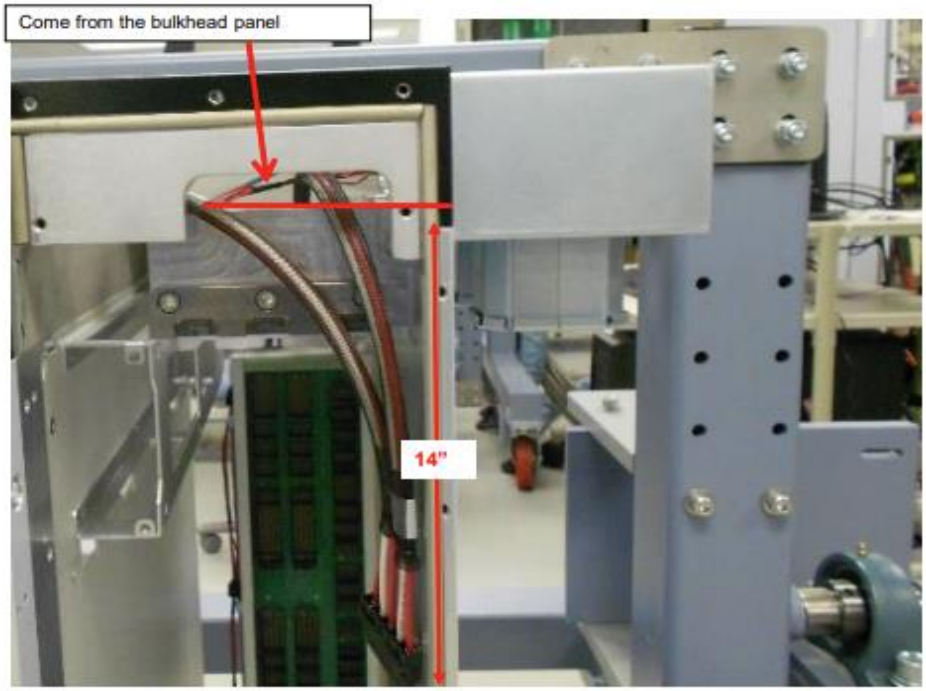
Measuring Tape

METHOD:

Continued from the previous page

c. As shown in the figure, insert the other end marked SB25 J48" through the front opening of the T/H.

d. As shown, leave the SB25 line with a drape length of approximately 14 inches.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

Measuring Tape

METHOD:

9.6 Take the SB25 line 650-705-01 along the EMO line to the circuit breaker area as shown in the figure.

9.7 Install the wire terminal marked "RET_4" on the power distribution block. Make sure that the wire terminal enters the upper opening of the power distribution block and is locked.

The tool is a flat-blade torque screwdriver with a torque value of 0.7 N•m.

9.8 Install the 650-705-01 wire terminal marked "CB_4" to the circuit breaker CB4 and lock it.

Note: Make sure the wire is inserted into the lower hole of the circuit breaker.

The tool is a PH2 Phillips head screwdriver with a torque value of 2.8 N•m.

9.9 Bind the SB25 cable and the adjacent cable with the cable tie that was already there

MAN:

CERTIFIED OPERATOR, TECHNICIAN

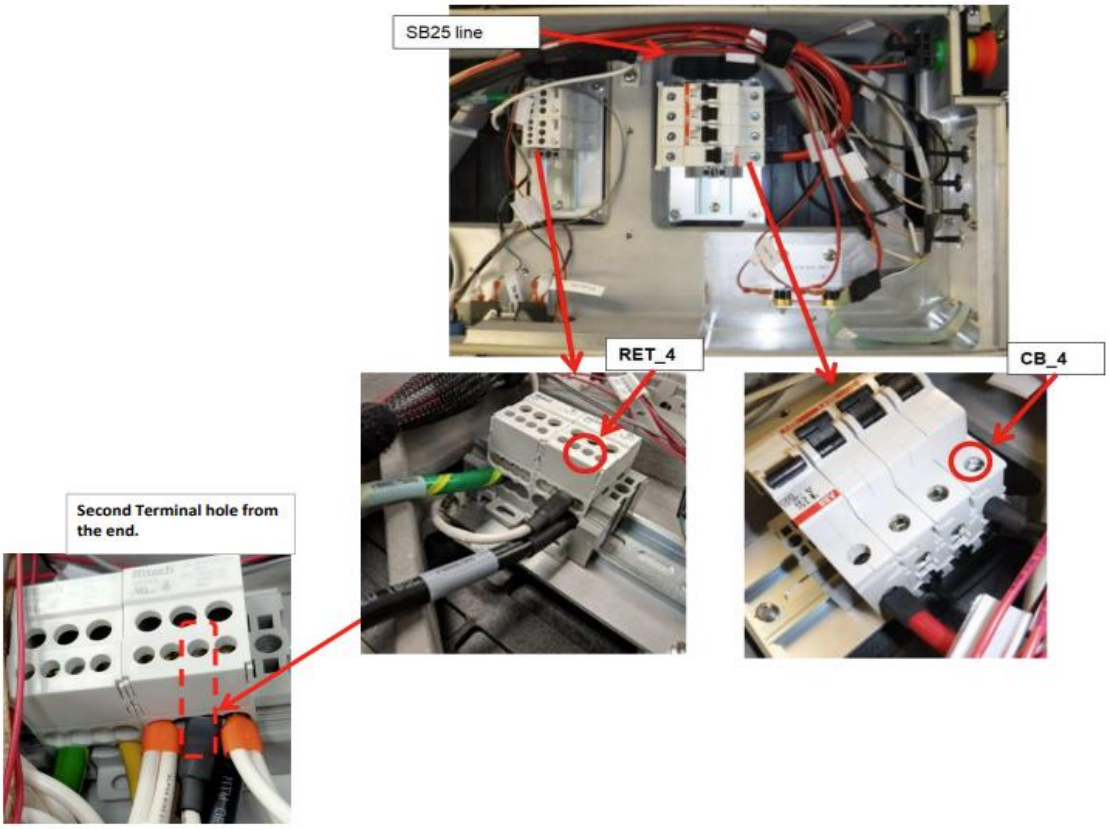
MATERIAL:

N/A

MACHINE:

N/A

METHOD:



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

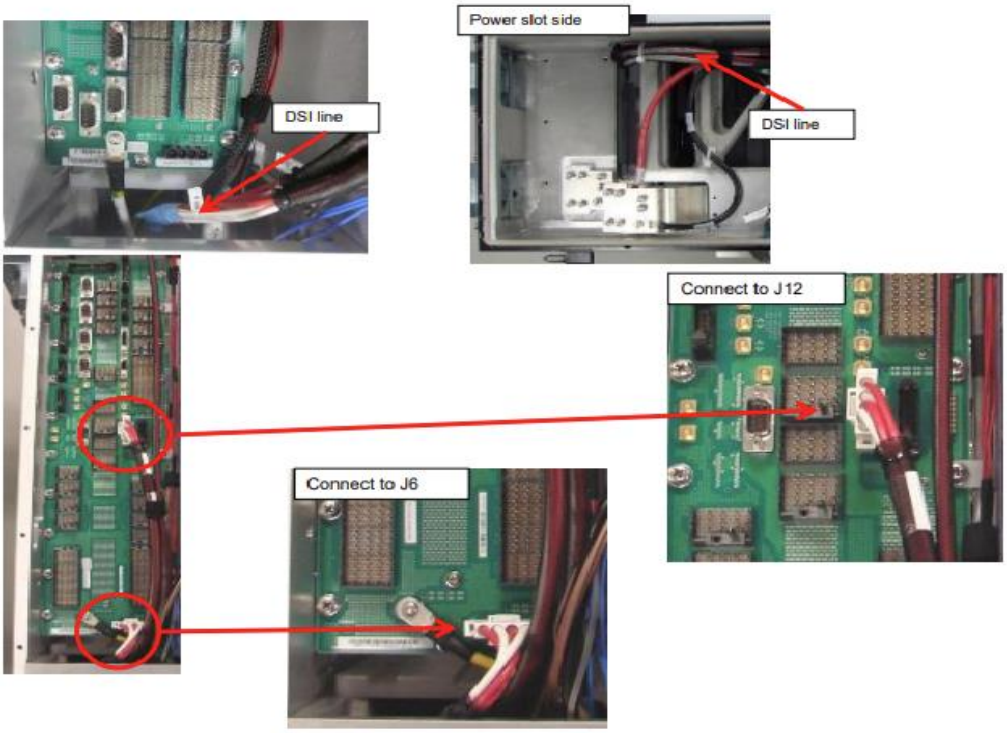
ITEM	PART NUMBER	DESCRIPTION	QTY
20	637-238-00	CABLE UF2 SPLIT DSI POWER AU CONTACT 48V TELECOM WITH CONN	1
21	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

paper tape

METHOD:

10.1 Take one DSI power cord 637-238-00 and wrap the thread with paper tape.
10.2 Pass the DSI line through the channel from the backplane side to the power slot side of the T/H.
10.3 Plug the J6 and J12 of the DSI cable into the connectors on the backplane.
10.4 Use one cable tie 813-929-01 to straighten and tie the DSI line with the EMO and SB25 lines.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

PH2 Phillips screwdriver 2.8 N•m, f

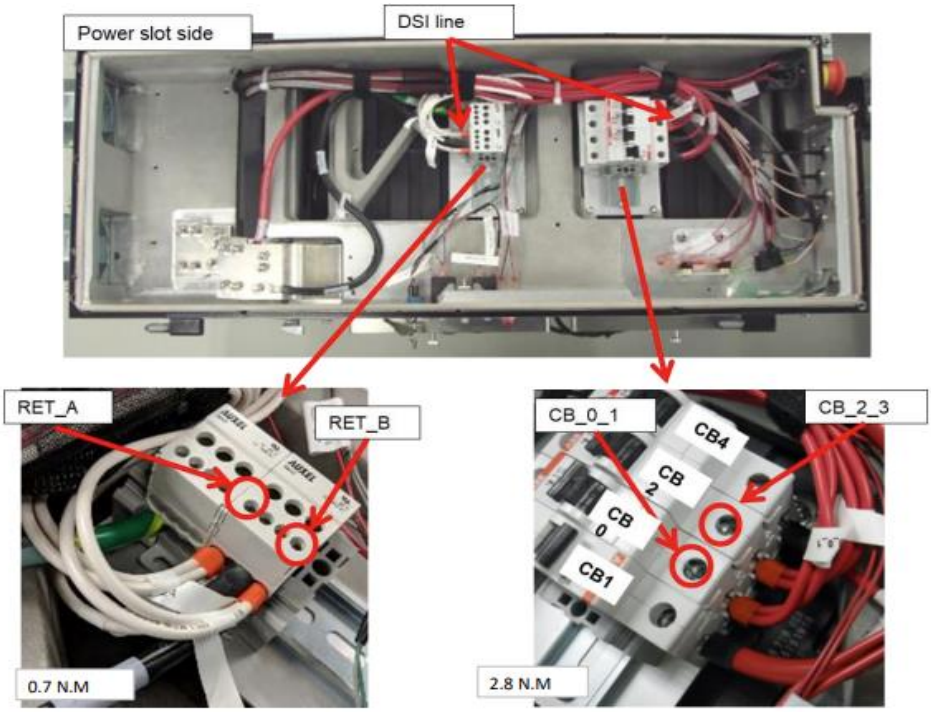
METHOD:

10.5 On the side of the power supply slot, pull out the DSI line and walk along the EMO line to the circuit breaker area.

10.6 Install the wire terminal marked CB_0_1 to the circuit breaker CB0, and the cable terminal marked CB_2_3 to the circuit breaker CB2.

10.7 Install the cable terminals marked RET_A and RET_B to the power distribution block. The tool is a flat-blade torque screwdriver with a torque value of 0.7 N•m.

10.8 As shown in the figure, put the excess cable into the T/H and use the existing cable ties to tie it with other cables.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

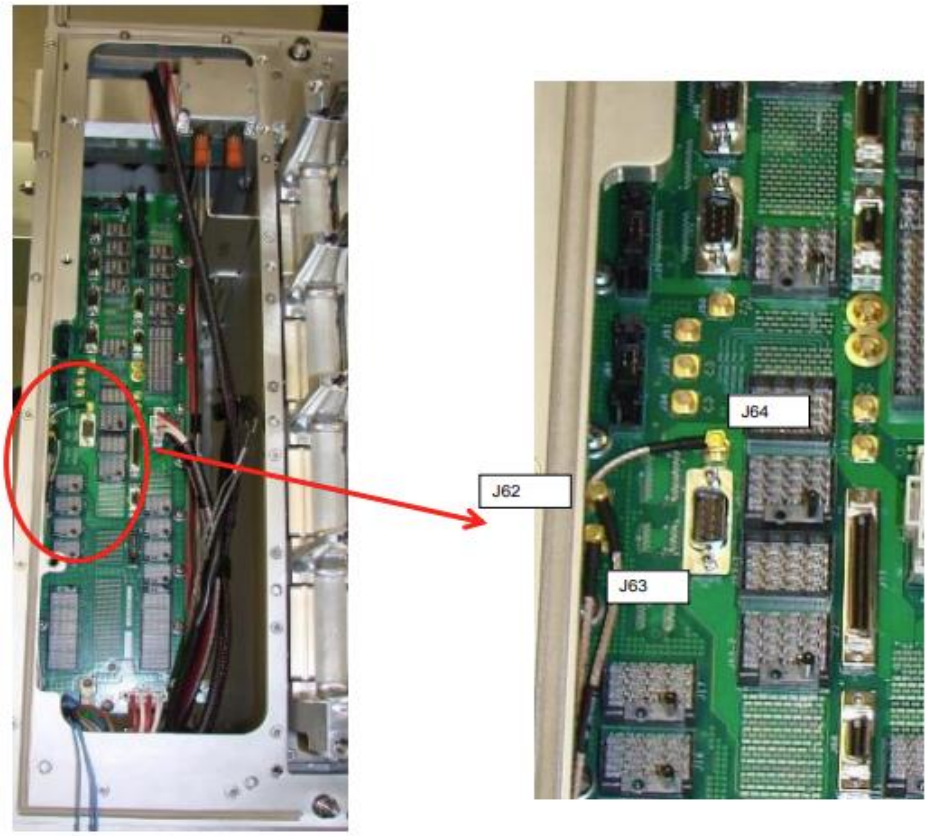
MACHINE:

N/A

METHOD:

11.1 Shake T/H to the vertical position as shown in the figure.

11.2 Insert the terminals marked "SUPPLANE J64", "SUPPLANE J63" and "SUPPLANE 62" (TH Existing hanging Scopesync Cables 627-090-02) on the PCBA boards J64, J63 and J62 respectively.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
22	938-104-01	4 INCH TIE WRAP BLA	1

MACHINE:

N/A

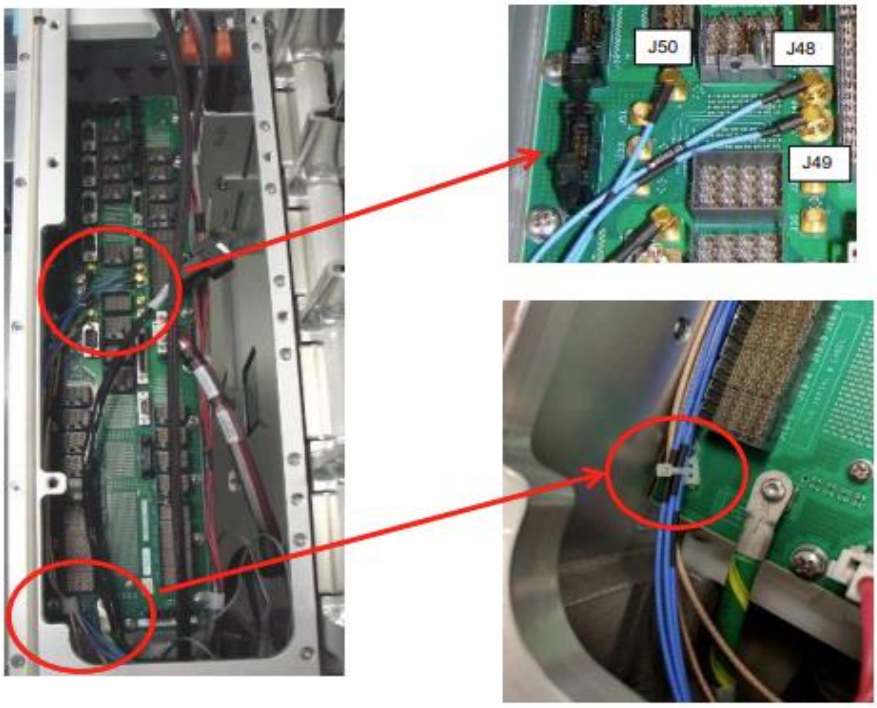
METHOD:

12.1 Lay the TDR cable along the inner wall of the T/H side near the PCBA position J48, J49, J50.

12.2 Install the cable ends marked "SUPPLANE J48", "SUPPLANE J49" and "SUPPLANE 50" on PCBA J48, J49 and J50 respectively.

12.3 As shown in the picture, use a cable tie 938-104-01 at the corner of the PCBA board to tie the wire, not too tight.

12.4 Organize all the cables and tie them together with the existing cable ties and EMO cables.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

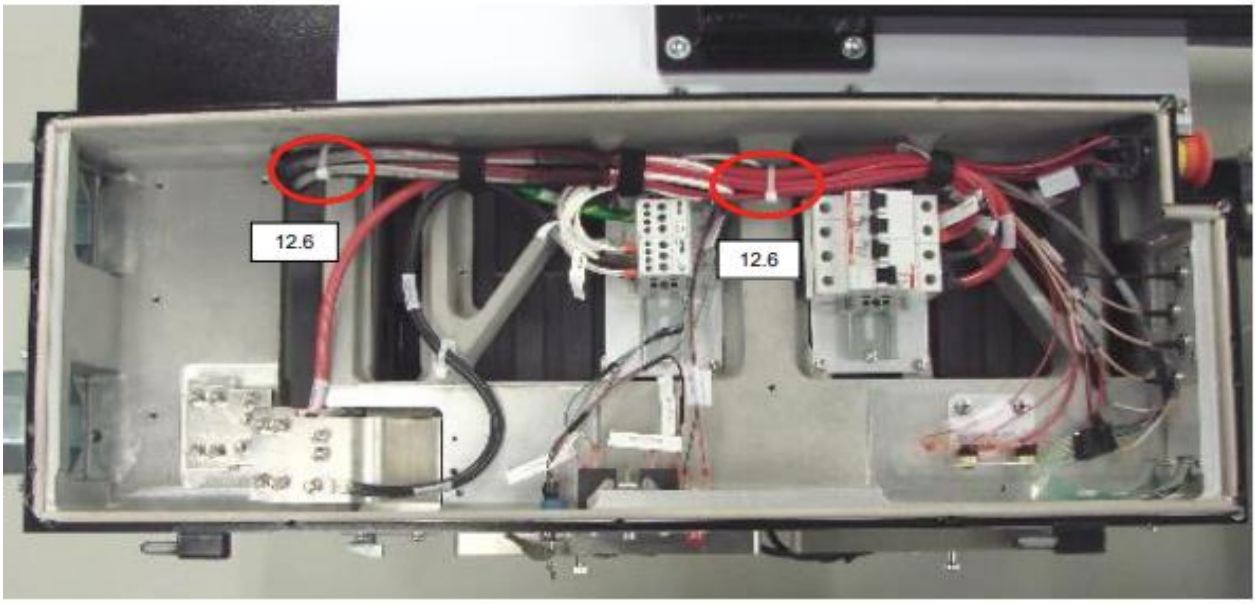
ITEM	PART NUMBER	DESCRIPTION	QTY
23	938-102-00	TIE WRAP 6.5 X 3/16"	2

MACHINE:

N/A

METHOD:

12.5 As shown in the figure, arrange all the wires and tie them together using the existing cable ties.
12.6 As shown in the figure, use two cable ties 938-102-00 to tie the cable.
12.7 Adjust the signal wires around the circuit breaker as needed to ensure that the signal wires have sufficient length to connect to the backplane.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
24	653-832-01	ELEC-MECH DB12 SUPPLY FLEX TUBE W/ QD ASSEMBLY FOR SEMI TEST	1

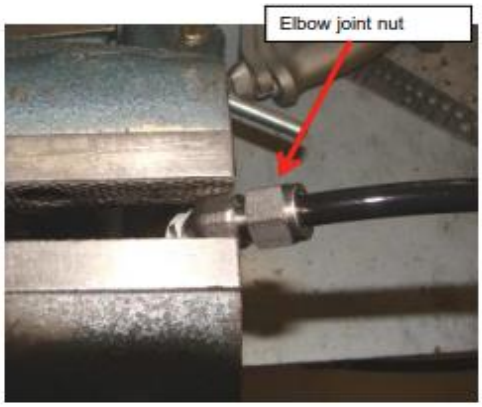
MACHINE:

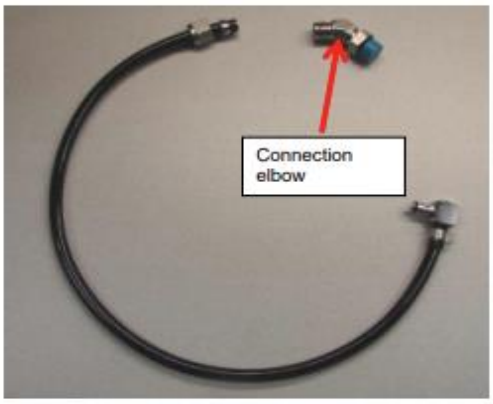
11/16” open-end wrench 27 N•m.

METHOD:

13.1 Take one QD assembly 653-832-01, place the QD assembly on the bench vise, and clamp it on the valve body of the elbow connector as shown.

13.2 Using an 11/16” open-end wrench, turn the union nut to loosen and remove the HFE hose.





MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
25	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1

MACHINE:

N/A

METHOD:

13.3 Remove the elbow fittings from the previous steps and remove the end cap.
13.4 Use Molykote grease compound 638-030-00 to lubricate threads and o-rings.
13.5 Retract the hexagon nut on the elbow fitting to the root of the elbow.

MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

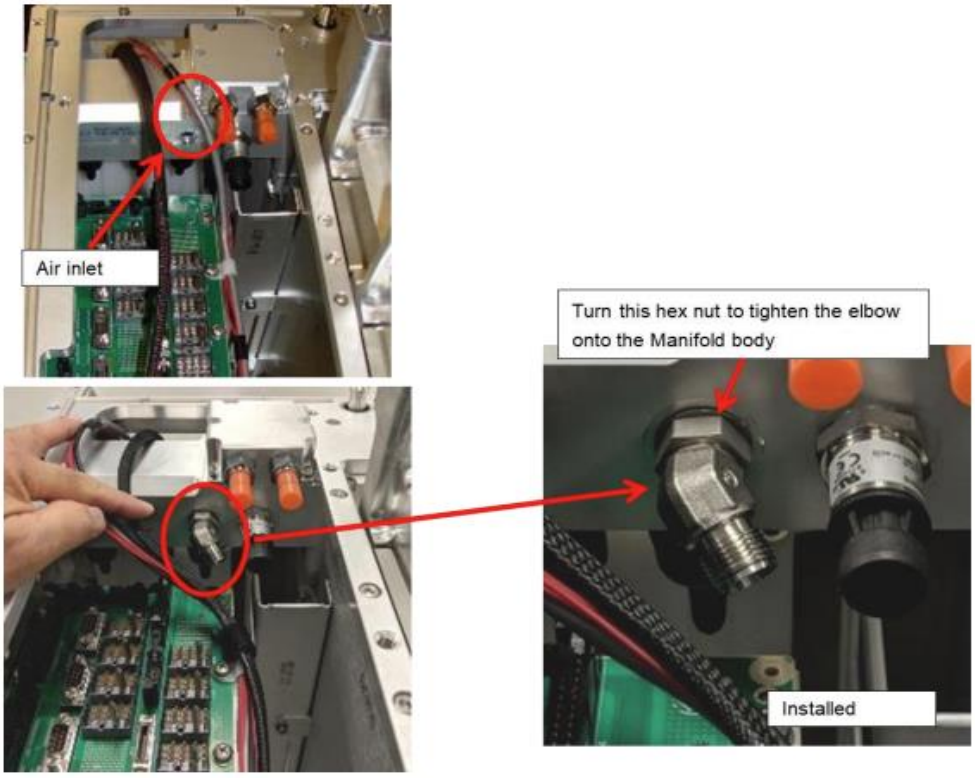
N/A

MACHINE:

11/16" open-end wrench 27 N•m.

METHOD:

13.6 Install the pipe joint on the air inlet on Manifold as shown in the figure and tighten it manually, then rotate the pipe joint in the opposite direction to the direction shown in the figure (reverse direction). The rotation must be less than one revolution). Use an 11/16" open-end wrench to tighten the hex nut with a torque value of 27 N•m.
Use 17mm to support pipe joint.



MAN:

CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

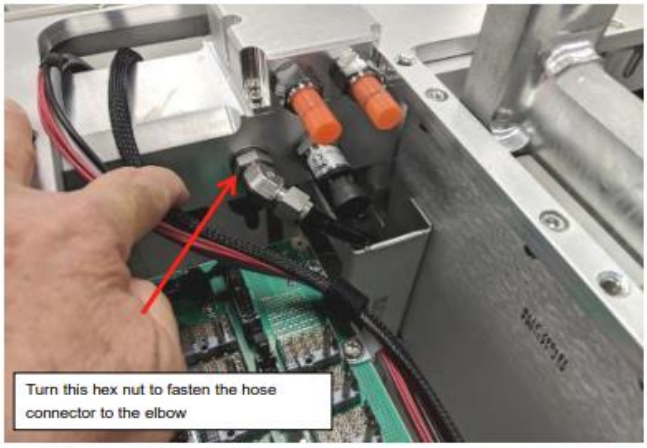
N/A

MACHINE:

marker, adjustable wrenches

METHOD:

13.7 Pass the end of the HFE hose connected to the elbow from the position shown to the side of the backplane.
13.8 Install one end of the hose to the elbow connector, push it all the way in and fix it completely.
13.9 Then use two wrenches 17mm dan19mm to fix the elbow and a hose hexagon nut to lock the hexagon nut, and mark with a marker after locking.
Then tighten the hex nut 1/4 turn.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Matters needing attention

1. Do ESD and protective safety measures.
2. Be careful during the assembly process so as not to scratch the surface of the parts.
3. After assembly, check the assembled parts for defects and leak-free operation.